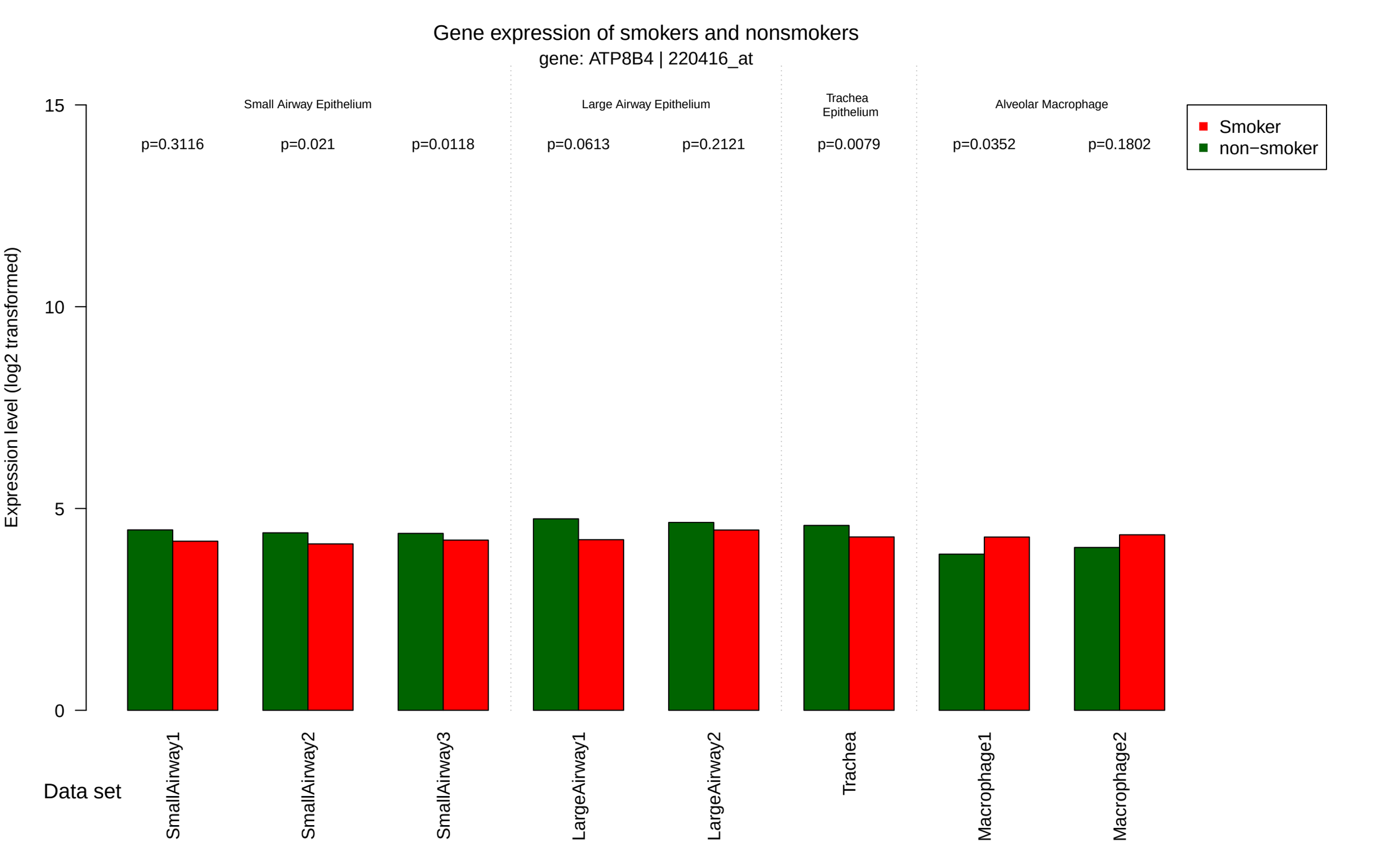
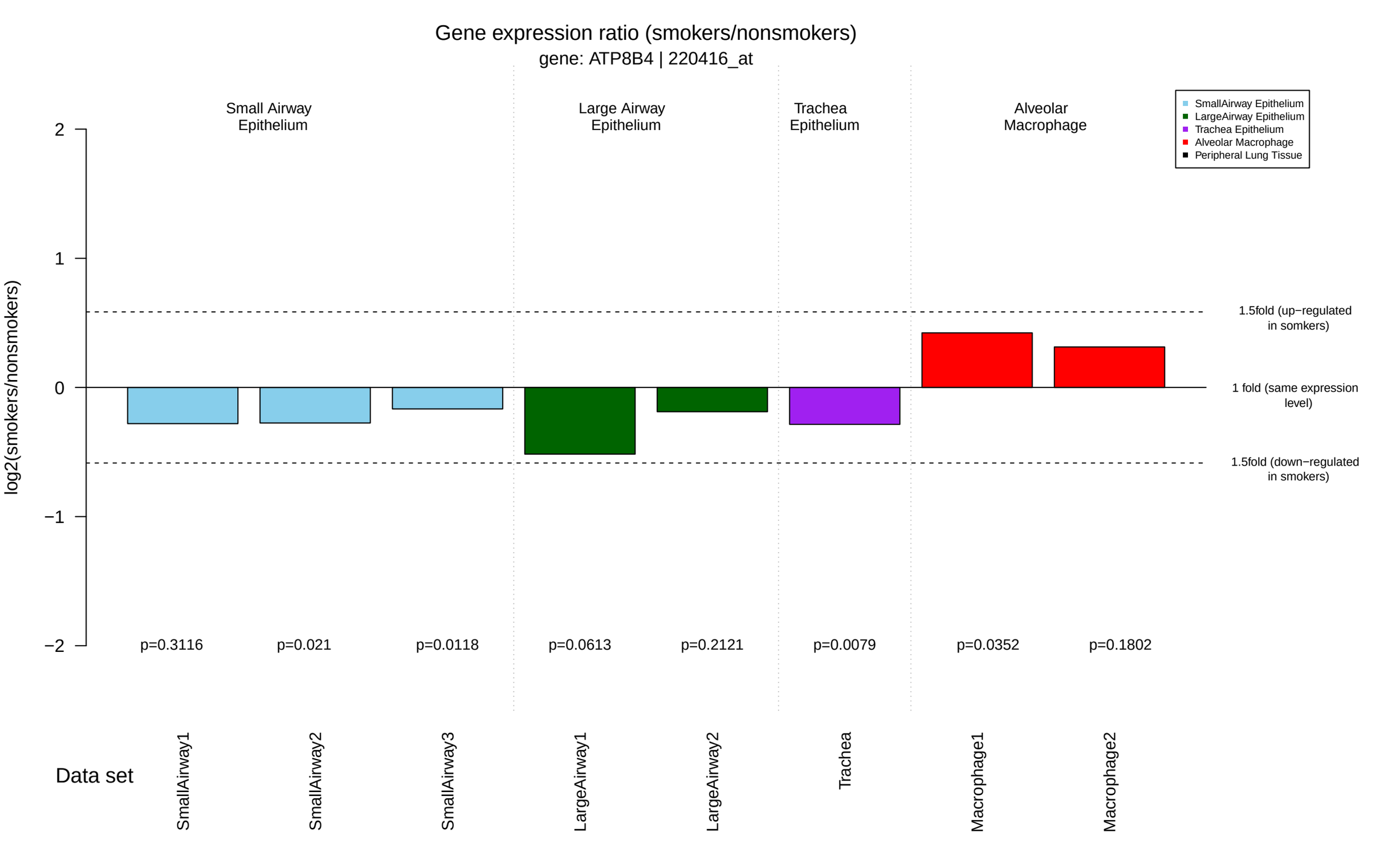
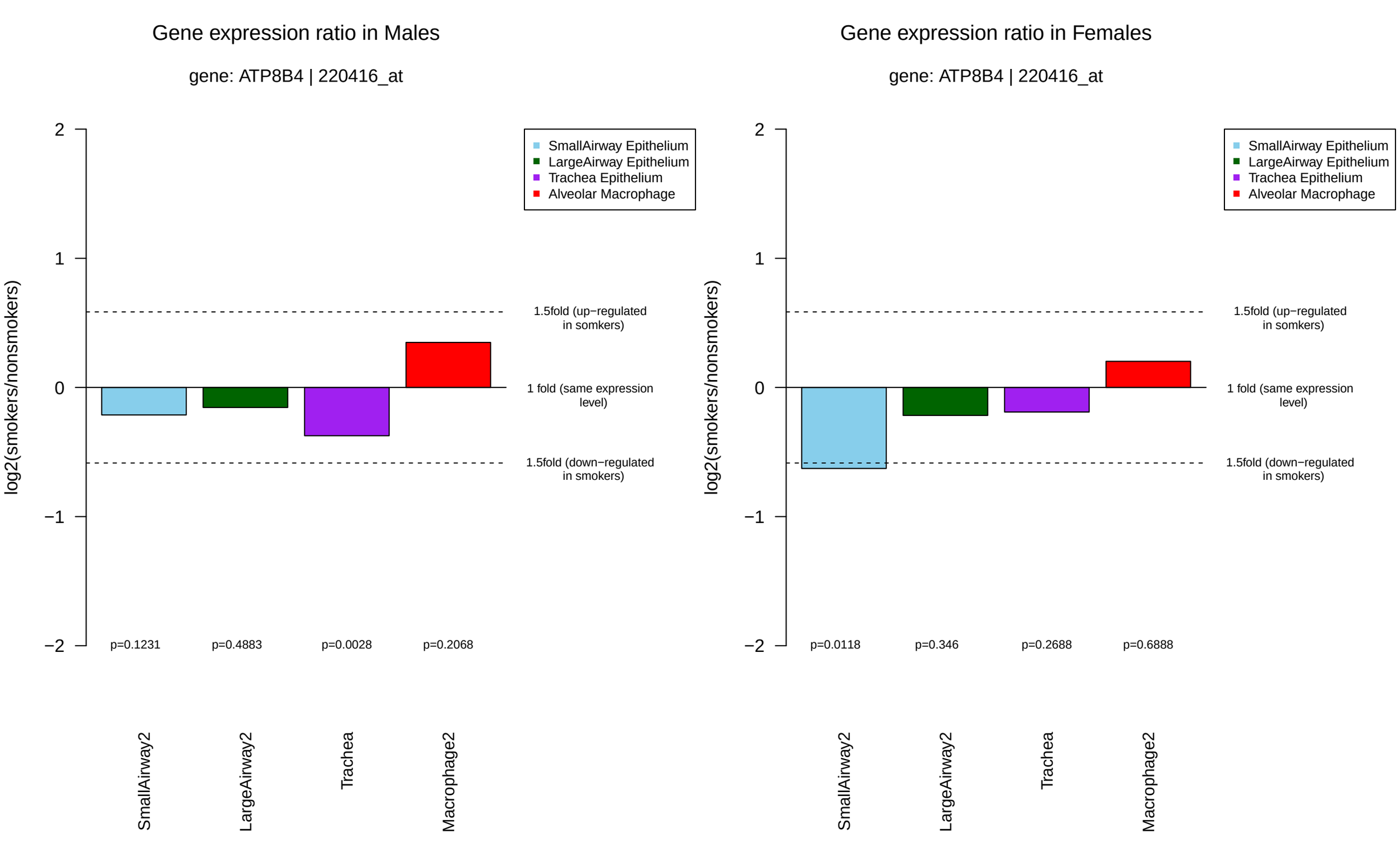
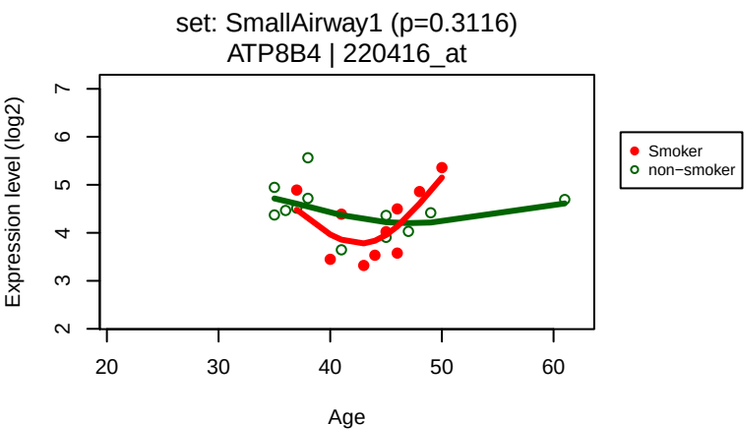
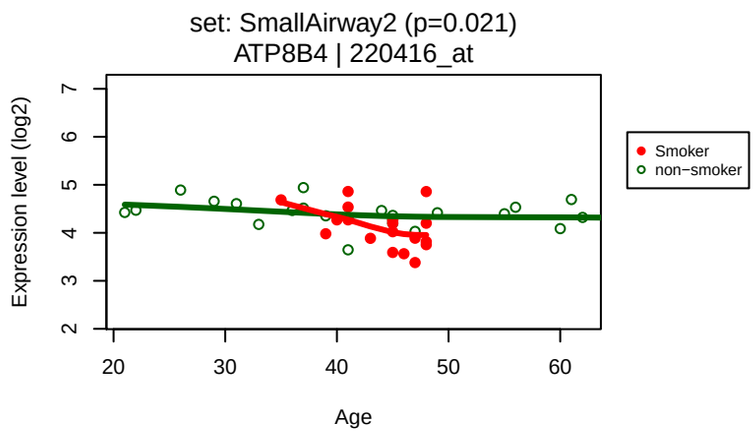
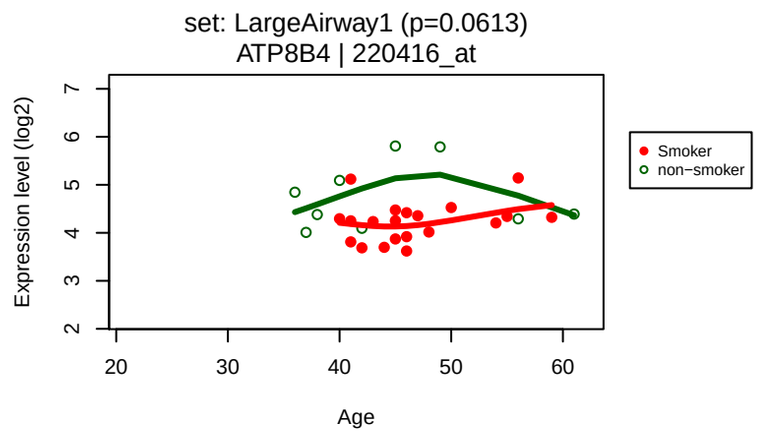

Gene expression of smokers and nonsmokers
gene: ATP8B4 | 220416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3116
p=0.021
p=0.0118
p=0.0613
p=0.2121
p=0.0079
p=0.0352
p=0.1802
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

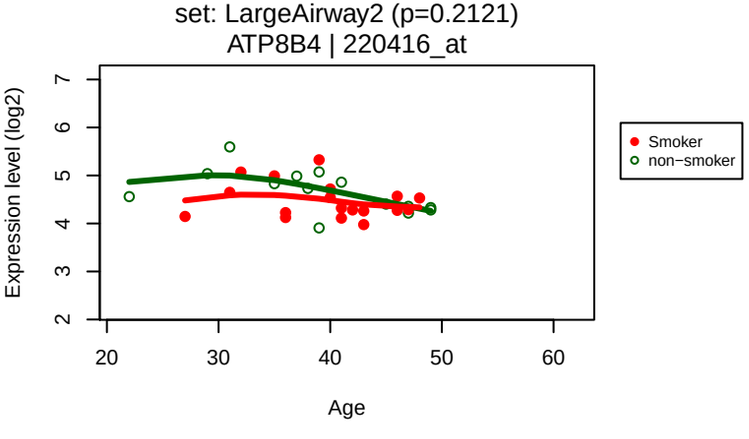

Gene expression ratio (smokers/nonsmokers)
gene: ATP8B4 | 220416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3116
p=0.021
p=0.0118
p=0.0613
p=0.2121
p=0.0079
p=0.0352
p=0.1802
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

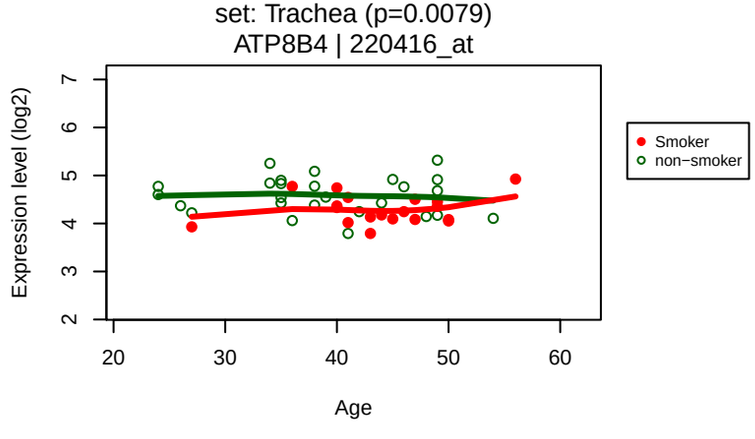

Gene expression ratio in Males
gene: ATP8B4 | 220416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1231
p=0.4883
p=0.0028
p=0.2068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8B4 | 220416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0118
p=0.346
p=0.2688
p=0.6888
SmallAirway2
LargeAirway2
Trachea
Macrophage2

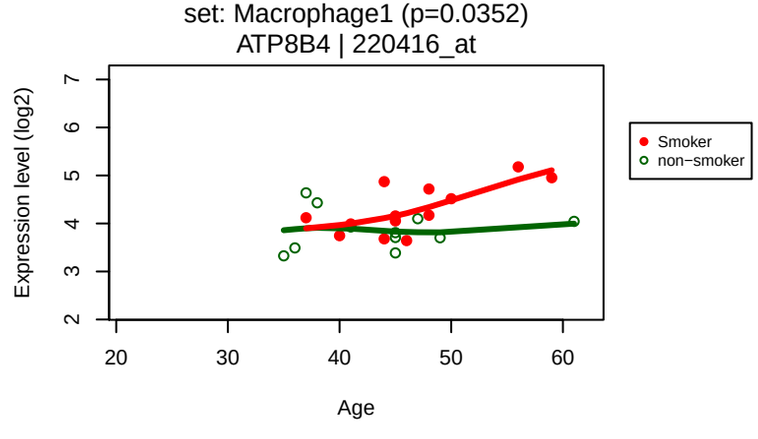

set: SmallAirway1 (p=0.3116)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker

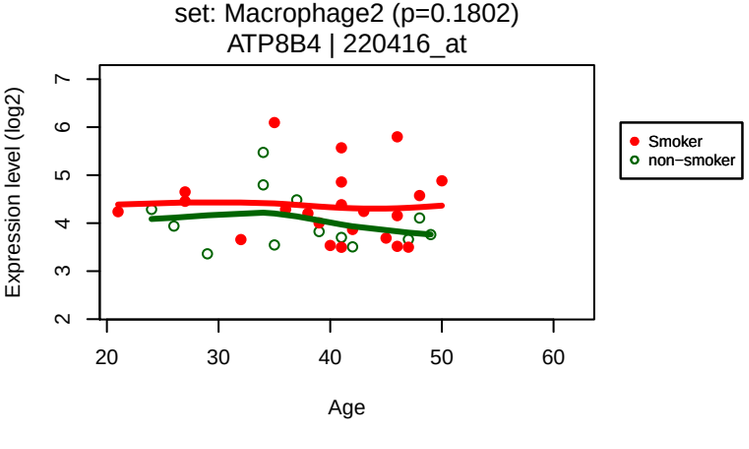

set: SmallAirway2 (p=0.021)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker

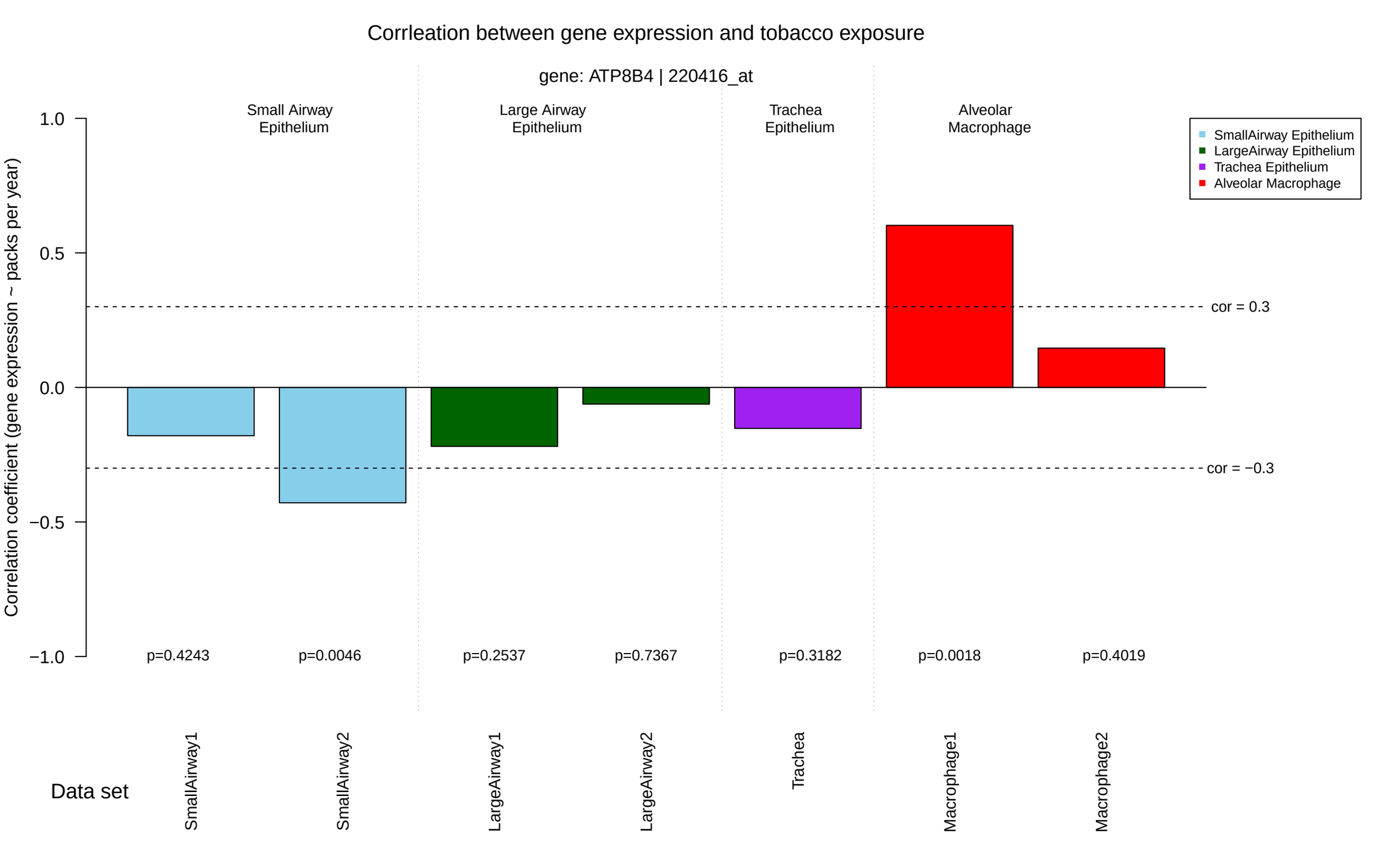

set: LargeAirway1 (p=0.0613)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2121)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0079)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0352)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1802)
ATP8B4 | 220416_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8B4 | 220416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4243
p=0.0046
p=0.2537
p=0.7367
p=0.3182
p=0.0018
p=0.4019
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2